

Tax calculation benchmark

Project description, expert instructions and taxonomy

Objective. Evaluate whether an LLM retrieves the correct official rules for a jurisdiction and period, applies them to the facts, and computes an auditable numeric answer with precise citations.

List of Countries. Kazakhstan: Kazakh/Russian; Pakistan: English; India: English; China: Chinese; Indonesia: Indonesian; Poland: Polish; Egypt: Arabic.

Assignment. (Stage 1) Each expert creates 25 distinct base scenarios, each with four additional variants: 125 cases. **(Stage 2)** Then each expert validates another expert's questions

Year coverage. Of each expert's 125 cases, target 75 in 2020–2026 (current laws), 25 in 2010–2019 (a bit older) and 25 in 2000–2009 (very old).

Quality. Use realistic fictional facts, interacting rules and dependent calculations. **DO NOT create simple lookup questions; the questions should be complex enough to include correct document retrieval for tax rules/laws and reasoning for multi-step calculations.** Exclude factual lookup, standalone deadlines, deliberately missing facts and one-step rate multiplication. Ask for one primary numeric result.

LLM-authored questions, solutions and translations are prohibited.

Build a family of five cases

Variant	How to create it
Original Template	Write one complete, independently solvable base case.
Numbers	Change amounts within approved ranges; keep the same legal conditions.
Threshold	Move below, at or above a bracket/cap; verify the applicable branch.
Eligibility	Change one relevant fact; verify the deduction, exemption or credit.
Year	Use another verified period; update rules, steps, answer and citations.
Missing Information	Intentionally remove an important information in the original question so that the question cannot be answered; include the details in gold steps.

Use Original Template once per family; choose the actual change type for the other four rows. **Types may repeat (or you can skip other variant types if they don't apply in your case)**. Change one factor at a time and record the change in Note. A different year requires verified rules and documents; never manufacture or edit statutory text.

Note: In the real world, people tend to skip important information when asking a tax-related question. To simulate that, you can make **one** of 4 variants of the same question family to intentionally have no answer because there is missing information, but **make sure to include what is missing in the final answer text and gold steps.**

Question Distribution: 6 tax categories, 4 original template questions per category, and an additional question of your choice (but still has to be in one of the categories) = overall 25 original template questions with 5 variants each, resulting in 125 questions in total. If you absolutely cannot create 5 variants per original question template, then create a new original question (26th, 27th, etc.), but make sure that the final number of questions is still 125.

2. Working Excel File

No / Family / Variant. Each case uses one row. Keep No unchanged. A Family has five cases, comprising one Original and four variants.

Country / Language. Prefilled from the country selected in Options. Write prose in each row's language. For Kazakhstan, fill shared metadata and Answer Value in the first language row; the paired row updates automatically. For single-language countries, complete the one row directly.

Tax category / Primary taxpayer. Select one of each. Use the Taxonomy tab. State relevant residence, registration and employment facts in Question; the broad taxpayer label does not replace these facts.

Year period (target) / Case Year. The bucket is suggested. Enter the transaction year or ending calendar year of the tax period. Specify exact dates and financial/assessment years in Question.

Difficulty. Choose by reasoning complexity, with no fixed quotas.

D2. Two or three connected legal decisions or calculations.

D3. Several dependent steps involving a cap, exception or interacting rules; the main target.

D4. Complex interaction of legal versions, jurisdictions or treaty rules with calculation. A different year alone does not make a case D4.

Question. Include all necessary facts, jurisdiction, period, currency and rounding. Each variant must stand alone; do not write "same as above."

Final Answer Text / Answer Value / Unit. Give a short answer and one numeric value. Unit defaults to the assigned currency; change it if needed (in the first language row for Kazakhstan). Zero is valid. Put intermediate amounts in Gold Steps; all final answers are numeric.

Gold Steps. Keep these labels and consecutive numbering. Use the question language for prose. Write numbers without thousands separators, using a decimal point. Cite each legal step and calculation rule. Leave Unit blank for nonnumeric results. Format:

Step 1: <legal decision or calculation>
 Result 1: <number or short conclusion>
 Unit 1: <unit; blank if not applicable>
 Citations 1: C1

Step 2: <next legal decision or calculation>
 Result 2: <number or short conclusion>
 Unit 2: <unit; blank if not applicable>
 Citations 2: C1, C2

Status / Note. Use Draft, Ready, Needs correction or Validated. Only the reviewer sets Validated. Record variant changes, source uncertainties and corrections in Note.

3. Evidence / References

Citations tab

Each Question No / Language pair has C1–C3. Use one row per provision/version. Delete unused rows; add C4, C5, etc. under the same pair when needed.

Question No / Language / Citation ID. Match No and Language to Tax Questions. Citation IDs match that row's Gold Steps and restart for each pair. Language identifies the question version.

Official title / Law number. Enter the original official title and identifier. Leave Law number blank when it does not apply.

Article / section / Paragraph / Subparagraph. Give the exact legal location; leave inapplicable fields blank.

Official source URL / Source file. Link to the official version; save the unchanged document in the Documents Folder and enter its filename. Complete citations for the assigned language, or both languages for Kazakhstan; reuse documents when applicable.

Supporting passage. Copy the exact text supporting the step. Keep official titles and quoted passages in the source language, including for translated questions.

Rule applies from / Rule applies until. Enter YYYY-MM-DD boundaries for the exact provision/version. These are not the Case Year or filing deadline; never assume January–December. Leave until blank for an ongoing rule. For unknown dates, leave blank, flag the citation in Note and resolve applicability before validation.

IMPORTANT NOTE on Official-source policy. Use official legislation, regulations, treaties, guidance, rate tables or decisions. Exclude commercial sites and blogs.

4. Taxonomy and coverage

The six categories follow OECD Revenue Statistics, an international classification by tax base.

OECD	Tax category	Calculation coverage
1000	Income, profits and capital gains	Personal/business income, gains, deductions, credits and income-tax withholding.
2000	Social security contributions	Compulsory contributions to qualifying government social-security schemes.
3000	Payroll and workforce taxes	Employer levies on wage bills or employee numbers.
4000	Property taxes	Ownership, wealth, inheritance, gifts and property/financial transaction taxes.
5000	Goods and services taxes	VAT/GST, sales/turnover taxes, excise, customs and taxes on use of goods.
6000	Other taxes	Residual taxes outside the preceding bases.

Check this link for detailed sub-categories of the taxonomy: [OECD classification of taxes and interpretative guide](#)